

## Chapter 3F

# Liver Disease, Cirrhosis, Portal Hypertension, and Hepatic Masses

Original study chapter for graph-RAG extraction and internal medicine board-style reasoning.

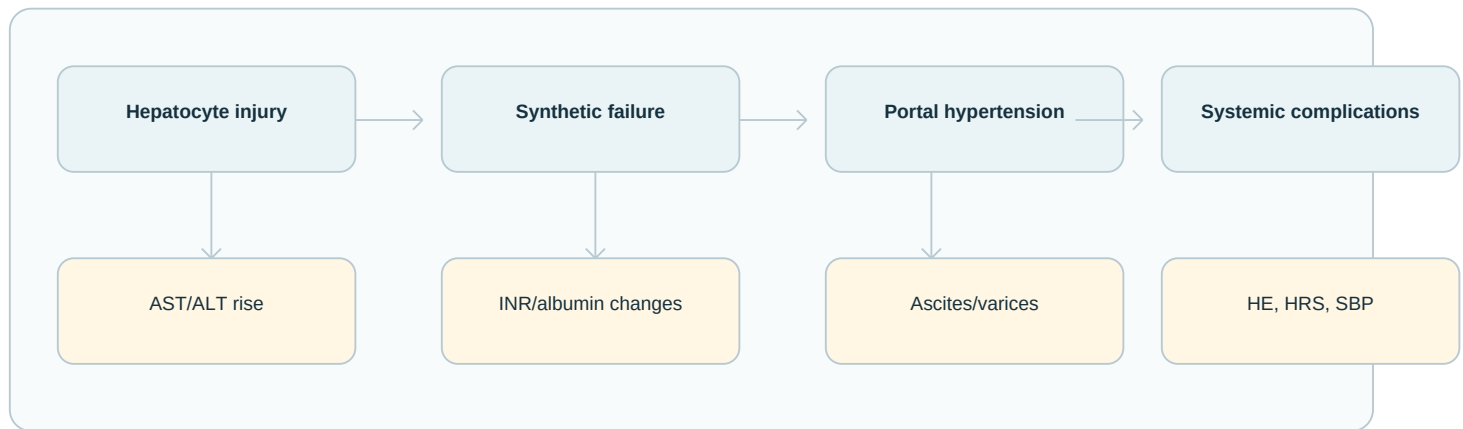
### Use note

This chapter is original Medvale writing. Uploaded reference pages were used only as topic coverage signals. It emphasizes concept relationships, numeric thresholds, illness scripts, diagnostic pathways, treatment priorities, and complications for future graph-RAG use.

## Chapter Roadmap

This liver chapter follows the GI overview and intestinal chapters by shifting from lumen-based disease to hepatobiliary synthetic function, portal hemodynamics, toxin metabolism, inherited storage disease, and liver masses. The high-yield clinical split is not simply hepatocellular versus cholestatic injury; it is whether the patient has acute liver failure, compensated chronic liver disease, decompensated cirrhosis, portal-hypertensive complications, or a focal hepatic lesion that may represent hepatocellular carcinoma.

- **Acute liver failure:** a rapid syndrome of severe hepatocyte injury with encephalopathy and impaired synthesis, usually defined by INR  $\geq 1.5$  in a patient without known cirrhosis and illness duration under about 26 weeks.
- **Cirrhosis:** chronic fibrosis with regenerative nodules that distorts blood flow, increases portal pressure, and reduces synthetic/metabolic reserve.
- **Portal hypertension:** the hemodynamic engine behind ascites, varices, splenomegaly, hypersplenism, hepatorenal syndrome, and portosystemic encephalopathy.
- **Inherited/metabolic liver disease:** Wilson disease and hereditary hemochromatosis are classic systemic diseases in which the liver is one of several damaged organs.
- **Hepatic masses:** hemangioma, focal nodular hyperplasia, hepatic adenoma, and hepatocellular carcinoma are distinguished by risk profile, imaging behavior, complications, and whether biopsy is safe or useful.



## 1. First Principles: What the Liver Does and How It Fails

The liver performs synthetic, detoxifying, immune, vascular, storage, and excretory tasks. These functions fail in different combinations. A patient with very high aminotransferases may have dramatic hepatocyte injury but still retain synthetic function if the INR and mental status are normal. Conversely, a patient with modest AST and ALT values may have advanced cirrhosis if albumin is low, platelets are low from hypersplenism, INR is prolonged, or ascites is present. For this reason, liver reasoning should separate injury markers from function markers.

| Signal                       | What it reflects                        | Typical examples                                                                                       | Graph-RAG relationship                                       |
|------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| AST and ALT                  | Hepatocellular injury, not reserve      | Viral hepatitis, ischemic hepatitis, acetaminophen toxicity, autoimmune hepatitis                      | Hepatocyte necrosis/inflammation -> aminotransferase release |
| Alkaline phosphatase and GGT | Bile duct injury or cholestasis         | Biliary obstruction, primary biliary cholangitis, primary sclerosing cholangitis, infiltrative disease | Cholestasis -> ALP/GGT elevation                             |
| Bilirubin                    | Excretion and conjugation burden        | Hemolysis, hepatocellular dysfunction, obstruction, sepsis                                             | Impaired handling of heme/bile -> jaundice                   |
| INR/PT                       | Synthetic function for clotting factors | Acute liver failure, decompensated cirrhosis, vitamin K deficiency, warfarin                           | Loss of hepatic synthesis -> coagulopathy                    |

| Signal                                                                                                                                                                                                                                                                                                                                                                                                   | What it reflects                     | Typical examples                                          | Graph-RAG relationship                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------|------------------------------------------------|
| Albumin                                                                                                                                                                                                                                                                                                                                                                                                  | Longer-term synthetic function       | Cirrhosis, nephrotic syndrome, inflammation, malnutrition | Low oncotic pressure -> edema/ascites          |
| Platelets                                                                                                                                                                                                                                                                                                                                                                                                | Portal HTN and splenic sequestration | Cirrhosis with splenomegaly                               | Portal HTN -> splenomegaly -> thrombocytopenia |
| <b>Numeric anchor: liver test interpretation</b>                                                                                                                                                                                                                                                                                                                                                         |                                      |                                                           |                                                |
| Aminotransferases above 1000 IU/L strongly suggest severe hepatocellular injury such as ischemic hepatitis, acute viral hepatitis, acetaminophen toxicity, or severe autoimmune/drug-induced injury. In contrast, mild-to-moderate AST/ALT elevation does not exclude cirrhosis. INR, bilirubin, albumin, platelets, and clinical decompensation often tell you more about prognosis than AST/ALT alone. |                                      |                                                           |                                                |

## 2. Approach to Abnormal Liver Tests

The first diagnostic move is to classify the pattern. A hepatocellular pattern is dominated by AST and ALT. A cholestatic pattern is dominated by alkaline phosphatase, GGT, and direct hyperbilirubinemia. A synthetic-failure pattern is dominated by INR elevation, low albumin, hypoglycemia, encephalopathy, or worsening renal function. Real patients often have mixed patterns; the pattern simply determines the safest next question.

| Pattern             | Dominant tests                                      | Common causes                                                                                                                      | Initial next steps                                                                                                    |
|---------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Hepatocellular      | AST/ALT high out of proportion to ALP               | Alcohol-associated hepatitis, viral hepatitis, ischemic hepatitis, acetaminophen, autoimmune hepatitis, NAFLD/NASH, Wilson disease | Medication/toxin review; hepatitis serologies; alcohol history; autoimmune tests if persistent; ultrasound if unclear |
| Cholestatic         | ALP/GGT high with direct bilirubin                  | Choledocholithiasis, malignancy, PSC, PBC, drug-induced cholestasis, infiltrative disease                                          | Confirm hepatic source with GGT; RUQ ultrasound; MRCP/ERCP if obstruction suspected                                   |
| Synthetic failure   | INR high, albumin low, hypoglycemia, encephalopathy | Acute liver failure, decompensated cirrhosis, severe sepsis/ischemia, advanced malignancy                                          | Assess mental status, INR, glucose, lactate, renal function; urgent hepatology/transplant center evaluation if ALF    |
| Portal hypertensive | Platelets low, splenomegaly, ascites, varices       | Cirrhosis, portal vein thrombosis, Budd-Chiari, schistosomiasis, right heart failure                                               | Ultrasound with Doppler; diagnostic paracentesis if ascites; variceal screening                                       |

A disproportionate AST:ALT ratio greater than 2:1 suggests alcohol-associated liver injury, especially if values are usually below several hundred rather than in the thousands. A very high alkaline phosphatase without GGT elevation raises the possibility of bone disease rather than hepatobiliary disease. Direct hyperbilirubinemia points toward hepatocellular excretion failure or obstruction; indirect hyperbilirubinemia points toward hemolysis, impaired conjugation, or Gilbert syndrome.

## 3. Acute Liver Failure

Acute liver failure is a medical emergency because the liver loses detoxification and synthetic capacity over days to weeks. The core clinical triad is new hepatic encephalopathy, impaired coagulation with INR at least 1.5, and no known chronic liver disease. The 26-week time frame is useful because it separates acute liver failure from chronic cirrhosis with acute decompensation. In practice, the clinician must simultaneously stabilize the patient, search for a reversible cause, and contact a transplant-capable center early.

### Common causes and illness scripts

| Cause                  | Classic clue                                                                         | Pathophysiology                                                     | Immediate action                                                     |
|------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------|
| Acetaminophen toxicity | Very high AST/ALT, possible low bilirubin early, overdose may be hidden or staggered | Toxic NAPQI metabolite depletes glutathione and injures hepatocytes | Start N-acetylcysteine if suspected; do not wait for perfect history |

| Cause                     | Classic clue                                                                 | Pathophysiology                              | Immediate action                                                      |
|---------------------------|------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------|
| Viral hepatitis           | HAV, HBV, HEV; travel, exposure, pregnancy risk for HEV                      | Immune-mediated hepatocyte injury            | Serologies; supportive care; antiviral therapy for selected HBV cases |
| Ischemic hepatitis        | Shock, heart failure, hypoxemia; dramatic AST/ALT rise                       | Low perfusion causes centrilobular necrosis  | Restore perfusion and oxygen delivery                                 |
| Drug-induced liver injury | New medication, herbal supplement, antiepileptic, antibiotic, immune therapy | Idiosyncratic or dose-related hepatotoxicity | Stop offending agent; consider NAC in selected non-acetaminophen ALF  |
| Autoimmune hepatitis      | Young or middle-aged patient, other autoimmune disease, high IgG             | Immune attack on hepatocytes                 | Autoimmune markers; specialist-directed steroids if likely            |
| Budd-Chiari syndrome      | Abdominal pain, hepatomegaly, ascites, thrombosis risk                       | Hepatic venous outflow obstruction           | Doppler imaging; anticoagulation and decompression strategy           |

## Recognition and stabilization

Patients may present with malaise, nausea, jaundice, coagulopathy, hypoglycemia, renal injury, infection, cerebral edema, or progressive confusion. Hypoglycemia occurs because hepatic glycogen storage and gluconeogenesis fail. Coagulopathy occurs because the liver cannot synthesize clotting factors. Encephalopathy reflects impaired detoxification of gut-derived and systemic neurotoxins and may worsen quickly.

- Check glucose immediately and repeatedly; treat hypoglycemia aggressively.
- Trend INR/PT, bilirubin, AST/ALT, alkaline phosphatase, lactate, creatinine, electrolytes, arterial/venous blood gas if ill, and mental status.
- Give N-acetylcysteine promptly when acetaminophen toxicity is possible or history is unreliable.
- Avoid sedatives when possible because they obscure encephalopathy and worsen airway risk.
- Consult hepatology/transplant services early; deterioration can be abrupt even if initial appearance is deceptively stable.

### Pitfall

Do not correct INR with plasma simply to make the number look better unless there is bleeding or an invasive procedure. The INR is a severity marker in acute liver failure and cirrhosis; unnecessary correction can obscure trends and add volume burden.

## 4. Cirrhosis: Fibrosis, Nodules, and Lost Reserve

Cirrhosis is the final common architecture of many chronic liver injuries. Fibrosis and regenerative nodules distort hepatic sinusoids and portal venous inflow. The liver becomes a high-resistance vascular bed, so portal pressure rises. At the same time, fewer functioning hepatocytes remain available for albumin synthesis, clotting factor production, ammonia metabolism, bile excretion, glucose regulation, and immune clearance. A patient may remain compensated for years, then become decompensated with ascites, variceal hemorrhage, hepatic encephalopathy, jaundice, spontaneous bacterial peritonitis, hepatorenal syndrome, or hepatocellular carcinoma.

### Major etiologies

| Etiology                                                        | Typical clues                                        | Mechanism of chronic injury                                        | High-yield associations                                 |
|-----------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------|
| Alcohol-associated liver disease                                | AST:ALT often >2, macrocytosis, alcohol use disorder | Acetaldehyde toxicity, oxidative stress, steatohepatitis, fibrosis | Mallory-Denk bodies; abstinence improves outcomes       |
| Metabolic dysfunction-associated steatotic liver disease / NASH | Obesity, diabetes, dyslipidemia, mild ALT elevation  | Insulin resistance -> steatosis -> inflammation/fibrosis           | Common cause of cirrhosis and HCC in metabolic syndrome |

| Etiology                    | Typical clues                                                               | Mechanism of chronic injury                               | High-yield associations                           |
|-----------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------|
| Chronic hepatitis B or C    | Risk exposures, positive serology, chronic inflammation                     | Viral persistence and immune-mediated injury              | HCC risk; HBV can cause HCC without cirrhosis     |
| Autoimmune hepatitis        | Female predominance, autoimmune disease, high IgG                           | Immune hepatocyte injury                                  | Steroid-responsive in many cases                  |
| PBC                         | Middle-aged woman, pruritus, cholestatic labs, antimitochondrial antibodies | Autoimmune small intrahepatic bile duct destruction       | Fatigue, xanthomas, osteoporosis                  |
| PSC                         | Cholestatic labs, IBD especially UC                                         | Inflammatory stricturing of intra/extrahepatic bile ducts | Cholangiocarcinoma risk; beading on MRCP          |
| Hemochromatosis/Wilson/A1AT | Family history or systemic findings                                         | Storage/protein misfolding injury                         | Screen relatives when inherited disease confirmed |

## Severity scores

Severity scoring translates complications into prognosis and transplant priority. Child-Pugh uses ascites, encephalopathy, bilirubin, albumin, and INR/PT. Scores of 5-6 are class A, 7-9 class B, and 10-15 class C. MELD-Na uses bilirubin, INR, creatinine, and sodium to estimate short-term mortality and prioritize transplant allocation. The clinical lesson is that renal dysfunction and hyponatremia in cirrhosis are not side issues; they often indicate advanced circulatory dysfunction.

| Child-Pugh element | 1 point   | 2 points      | 3 points          |
|--------------------|-----------|---------------|-------------------|
| Ascites            | None      | Mild/moderate | Severe/refractory |
| Encephalopathy     | None      | Grade I-II    | Grade III-IV      |
| Bilirubin          | <2 mg/dL  | 2-3 mg/dL     | >3 mg/dL          |
| Albumin            | >3.5 g/dL | 2.8-3.5 g/dL  | <2.8 g/dL         |
| INR                | <1.7      | 1.7-2.3       | >2.3              |

## 5. Portal Hypertension and Variceal Bleeding

Portal hypertension arises when blood entering the liver through the portal vein encounters increased resistance. In cirrhosis, resistance comes from fibrotic architecture, contracted sinusoids, and dysregulated intrahepatic vascular tone. Blood then seeks collateral pathways, producing esophageal/gastric varices, caput medusae, hemorrhoidal collaterals, and splenomegaly. Variceal bleeding is dangerous because it can be brisk, recurrent, and superimposed on coagulopathy, thrombocytopenia, infection risk, and poor hepatic reserve.

### Acute variceal hemorrhage: stabilize, suppress portal flow, treat infection, band

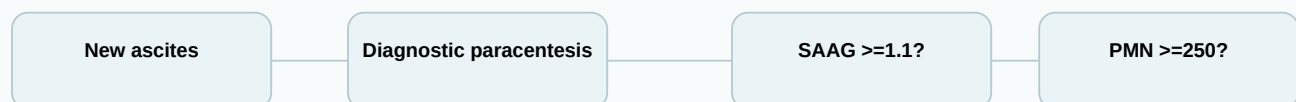
- **Airway and circulation:** assess mental status and aspiration risk; place large-bore IVs; resuscitate while avoiding unnecessary overtransfusion.
- **Restrictive transfusion concept:** many protocols target hemoglobin around 7 g/dL in stable patients, with higher targets for shock or active cardiac ischemia.
- **Vasoactive therapy:** octreotide is commonly used to reduce splanchnic inflow; terlipressin is used in many settings where available.
- **Antibiotics:** cirrhotic GI bleeding increases bacterial translocation and infection risk; ceftriaxone is commonly used in hospitalized variceal bleeding.
- **Endoscopy:** endoscopic variceal ligation is the usual first-line endoscopic therapy for esophageal varices once stabilized.
- **Rescue therapy:** balloon tamponade or self-expanding esophageal stents may temporize catastrophic bleeding; TIPS is considered for refractory or high-risk bleeding.

| Scenario                        | Primary strategy                                                                                                        | Secondary prevention                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Large varices, no prior bleed   | Nonselective beta blocker such as carvedilol/propranolol/nadolol or endoscopic ligation depending on risk and tolerance | Ongoing surveillance and portal pressure reduction       |
| Acute esophageal variceal bleed | Resuscitation, vasoactive drug, antibiotics, urgent endoscopic ligation                                                 | Nonselective beta blocker plus repeat band ligation      |
| Refractory/recurrent bleeding   | Evaluate for TIPS and transplant candidacy                                                                              | Manage encephalopathy risk and cardiac contraindications |
| Gastric varices                 | Often require specialized endoscopic glue/coiling, TIPS, or BRTO-type strategies                                        | Specialist-directed plan                                 |

## 6. Ascites, Spontaneous Bacterial Peritonitis, and Hepatorenal Syndrome

Ascites is the most visible sign of decompensated portal hypertension. The mechanism combines sinusoidal portal hypertension, arterial vasodilation, renal sodium retention, hyperaldosteronism, hypoalbuminemia, and impaired lymphatic drainage. New ascites should be tapped because physical exam cannot reliably separate uncomplicated portal-hypertensive ascites from infection, malignancy, pancreatitis, tuberculosis, or cardiac ascites.

### Ascites diagnostic logic



Yes: portal HTN likely. No: malignancy, infection, pancreatitis, nephrotic, etc.

Yes: treat SBP now; send culture before antibiotics

| Ascitic fluid result                 | Interpretation                                                       | Action                                                                                     |
|--------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| SAAG $\geq 1.1$ g/dL                 | Portal hypertension likely                                           | Think cirrhosis, alcoholic hepatitis, cardiac ascites, Budd-Chiari, portal vein thrombosis |
| SAAG $< 1.1$ g/dL                    | Portal hypertension less likely                                      | Think malignancy, TB, pancreatitis, nephrotic syndrome, serositis                          |
| PMN $\geq 250$ cells/mm <sup>3</sup> | Spontaneous bacterial peritonitis until proven otherwise             | Start antibiotics promptly; send culture; give albumin in higher-risk cases                |
| Low total protein                    | Reduced opsonic activity and higher SBP risk                         | May influence prophylaxis decisions in selected high-risk patients                         |
| Very high amylase                    | Pancreatic ascites or gut perforation                                | Image pancreas/bowel and treat source                                                      |
| Bloody fluid                         | Traumatic tap, malignancy, HCC rupture, TB, pancreatitis, infarction | Interpret with RBC correction and clinical context                                         |

### Management of uncomplicated ascites

First-line ascites management combines sodium restriction, diuretics, and monitoring. Spironolactone targets aldosterone-mediated sodium retention; furosemide helps balance potassium and augment natriuresis. Large-volume paracentesis relieves tense ascites, and albumin is commonly given when large volumes are removed to prevent circulatory dysfunction. Refractory ascites prompts evaluation for TIPS or transplant, but TIPS can worsen encephalopathy and is not suitable for every patient.

## Spontaneous bacterial peritonitis

SBP is infection of ascitic fluid without a surgically treatable intra-abdominal source. It can present with fever, abdominal pain, vomiting, diarrhea, worsening encephalopathy, renal injury, hypotension, or only subtle clinical decline. Diagnostic paracentesis is the key test. A PMN count of at least 250 cells/mm<sup>3</sup> is enough to treat, even if culture is negative. Culture-negative neutrocytic ascites behaves clinically like SBP.

### SBP anchor

In any patient with cirrhosis and ascites who is hospitalized, febrile, encephalopathic, hypotensive, renally worse, or has GI bleeding, diagnostic paracentesis should be strongly considered early. Delay increases mortality.

## Hepatorenal syndrome

Hepatorenal syndrome is functional renal failure caused by advanced cirrhotic circulatory dysfunction rather than intrinsic structural kidney disease. Splanchnic vasodilation lowers effective arterial blood volume, triggering intense renal vasoconstriction. The urine sodium is often very low, and bland urine sediment supports the diagnosis, but HRS is a diagnosis made after excluding shock, nephrotoxins, obstruction, and intrinsic kidney disease. Treatment typically combines albumin with vasoconstrictor therapy and urgent transplant evaluation.

## 7. Hepatic Encephalopathy

Hepatic encephalopathy is a neuropsychiatric syndrome caused by impaired hepatic detoxification and portosystemic shunting. Ammonia is important, but the diagnosis is clinical; a normal or elevated ammonia level alone should not replace bedside mental status assessment. Common precipitants include GI bleeding, infection, constipation, dehydration, excess diuresis, hypokalemia, sedatives, opioids, renal failure, high protein load in selected patients, and TIPS.

| Finding                                     | Meaning                                      | Management association                                                             |
|---------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------|
| Sleep-wake reversal, inattention, confusion | Early encephalopathy                         | Search for trigger; begin bowel-directed therapy                                   |
| Asterixis                                   | Motor sign of metabolic encephalopathy       | Suggests clinically meaningful encephalopathy but is not specific to liver disease |
| Somnolence, stupor, coma                    | Severe encephalopathy                        | Airway risk; ICU-level monitoring may be needed                                    |
| GI bleed or infection                       | Nitrogen load and inflammatory trigger       | Treat source, antibiotics when indicated, endoscopy if bleeding                    |
| Constipation                                | Increased intestinal ammonia absorption time | Lactulose titrated to about 2-3 soft stools/day                                    |
| Recurrent episodes                          | High relapse risk                            | Rifaximin is commonly added to lactulose in recurrent disease                      |

### Pitfall

Do not attribute every altered cirrhotic patient to hepatic encephalopathy. Infection, intracranial bleeding, alcohol withdrawal, hypoglycemia, hyponatremia, sedatives, and sepsis can mimic or precipitate encephalopathy and require separate treatment.

## 8. Inherited and Metabolic Liver Diseases

### Wilson disease

Wilson disease is an autosomal recessive disorder of copper metabolism caused by ATP7B dysfunction. Impaired biliary copper excretion and impaired incorporation into ceruloplasmin lead to copper accumulation in liver, brain, cornea, kidney, and other tissues. It often presents between childhood and young adulthood, but adult presentations occur. Clinical patterns include hepatitis, cirrhosis, acute liver failure, neuropsychiatric changes, movement disorder, hemolysis, renal tubular dysfunction, and Kayser-Fleischer rings.

| Wilson clue                                    | Interpretation                                                                                                                  |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Low ceruloplasmin                              | Supports diagnosis, but inflammation and pregnancy can confound levels                                                          |
| High 24-hour urinary copper                    | Reflects excess copper excretion and is useful diagnostically                                                                   |
| Kayser-Fleischer rings                         | Corneal copper deposition; especially common with neurologic disease                                                            |
| Low alkaline phosphatase in severe ALF pattern | Classic clue in fulminant Wilson presentations                                                                                  |
| Treatment                                      | Chelation such as penicillamine or trientine, zinc to reduce absorption, transplant for fulminant disease or refractory failure |

## Hereditary hemochromatosis

Hereditary hemochromatosis is most often an HFE-associated iron overload disorder. Increased intestinal iron absorption leads to iron deposition in liver, pancreas, heart, joints, pituitary-gonadal axis, and skin. Many patients are asymptomatic early. Later features include fatigue, arthralgias, diabetes mellitus, cardiomyopathy or arrhythmias, hypogonadism, hyperpigmentation, cirrhosis, and markedly increased HCC risk once cirrhosis develops.

| Test/feature                | High-yield interpretation                                                                             |
|-----------------------------|-------------------------------------------------------------------------------------------------------|
| Transferrin saturation >45% | Common screening threshold suggesting iron overload                                                   |
| High ferritin               | Reflects iron stores but also rises with inflammation, alcohol use, infection, and metabolic syndrome |
| HFE genetic testing         | Confirms common inherited forms when iron studies suggest disease                                     |
| Phlebotomy                  | Main treatment; removes iron and prevents complications if started early                              |
| Family screening            | Important because early treatment prevents irreversible organ injury                                  |

## 9. Benign and Malignant Liver Masses

Liver lesions are often found incidentally. The internal medicine task is to identify the danger pattern: a lesion in a cirrhotic liver, a bleeding-prone adenoma, a vascular hemangioma that should not be biopsied, or a benign focal nodular hyperplasia that can usually be observed. The same size lesion has different implications in a healthy young woman, a patient on anabolic steroids, and a patient with hepatitis B cirrhosis.

### Solid liver lesion pattern recognition

**Hemangioma**  
very common, vascular  
no biopsy

**FNH**  
central scar  
no malignant potential

**Adenoma**  
OCP/anabolic steroid  
bleed risk if >5 cm

**HCC**  
cirrhosis/HBV/HCV  
arterial enhancement

| Lesion                    | Risk group/setting                                            | Key concept                                                         | Management logic                                               |
|---------------------------|---------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------|
| Cavernous hemangioma      | Common incidental lesion; may enlarge with estrogen/pregnancy | Most common benign liver tumor; vascular                            | Usually observe; avoid biopsy because of bleeding risk         |
| Focal nodular hyperplasia | Women of reproductive age                                     | Hyperplastic response to vascular anomaly; central scar may be seen | Usually no treatment if confidently diagnosed and asymptomatic |



| Lesion                   | Risk group/setting                                                    | Key concept                                                                      | Management logic                                                                                                                     |
|--------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Hepatic adenoma          | OCPs, anabolic steroids, glycogen storage disease, metabolic syndrome | Benign but can bleed; small malignant potential, especially beta-catenin subtype | Stop hormones; resect or treat if large (often >5 cm), growing, male patient, symptomatic, or high-risk subtype                      |
| Hepatocellular carcinoma | Cirrhosis, chronic HBV/HCV, aflatoxin, hemochromatosis, NASH          | Malignant tumor often diagnosed by multiphasic imaging in at-risk patients       | Stage by liver function and tumor burden; options include resection, ablation, transplant, TACE, radioembolization, systemic therapy |

## 10. Hepatocellular Carcinoma

Hepatocellular carcinoma is strongly linked to chronic liver injury and cirrhosis, although chronic hepatitis B can cause HCC even without established cirrhosis. HCC may be asymptomatic, present with weight loss or right upper quadrant pain, appear as decompensation in a previously stable cirrhotic patient, or be discovered through surveillance. Paraneoplastic syndromes can include erythrocytosis, hypercalcemia, hypoglycemia, and diarrhea, but these are not required for diagnosis.

- **Surveillance:** high-risk cirrhotic patients are commonly screened every 6 months with ultrasound, often with AFP depending on local protocol and guideline version.
- **Diagnostic imaging:** multiphasic CT or MRI can show arterial phase hyperenhancement with portal/delayed washout in at-risk patients.
- **AFP:** useful as an adjunct and for monitoring in some patients, but normal AFP does not exclude HCC and elevated AFP is not perfectly specific.
- **Curative-intent options:** resection, local ablation, or liver transplant depending on tumor burden, portal hypertension, liver function, and performance status.
- **Locoregional/systemic options:** TACE, radioembolization, external beam strategies, and systemic immunotherapy/targeted therapy are used when curative options are not suitable.

### Clinical reasoning anchor

A new liver mass in a patient with cirrhosis should be treated as HCC until proven otherwise. A normal AFP does not safely rule it out. The next step is usually multiphasic CT or MRI rather than blind biopsy.

## 11. Integrated Diagnostic Algorithms

### Algorithm A: Jaundice

| Step | Question                                   | Interpretation                                                                                                                           |
|------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Is bilirubin mostly indirect or direct?    | Indirect suggests hemolysis, impaired uptake/conjugation, or Gilbert; direct suggests hepatocellular disease or cholestasis/obstruction. |
| 2    | Are AST/ALT or ALP/GGT dominant?           | AST/ALT dominant = hepatocellular; ALP/GGT dominant = cholestatic.                                                                       |
| 3    | Is there pain, fever, or ductal dilation?  | RUQ pain/fever/dilated ducts suggest obstruction/cholangitis; urgent biliary imaging and possible ERCP.                                  |
| 4    | Is INR elevated or mental status abnormal? | Think acute liver failure or decompensated cirrhosis; stabilize and consult urgently.                                                    |

### Algorithm B: Decompensated cirrhosis

When a cirrhotic patient becomes acutely ill, look for the precipitant rather than treating each complication as isolated. GI bleeding can trigger encephalopathy and SBP. SBP can trigger renal failure and encephalopathy. Excess diuresis can trigger hypokalemia, renal hypoperfusion, and encephalopathy. A practical admission bundle includes CBC, CMP, INR, cultures if

infectious concern, diagnostic paracentesis if ascites, medication review, renal evaluation, and careful search for bleeding, infection, thrombosis, alcohol use, and new malignancy.

| Presentation         | Must-not-miss diagnoses                                                                              | Early actions                                                                                                         |
|----------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Confusion            | HE, infection/SBP, GI bleed, hypoglycemia, hyponatremia, intracranial bleed, intoxication/withdrawal | Glucose, infection workup, paracentesis if ascites, lactulose, avoid sedatives                                        |
| Abdominal distention | Ascites, SBP, portal vein thrombosis, malignancy, cardiac ascites                                    | Diagnostic paracentesis; ultrasound/Doppler if unclear                                                                |
| Hematemesis/melena   | Variceal bleed, peptic ulcer, portal hypertensive gastropathy                                        | Resuscitate, octreotide, antibiotics, PPI if ulcer possible, urgent endoscopy                                         |
| Rising creatinine    | HRS, prerenal azotemia, ATN, nephrotoxin, obstruction                                                | Stop nephrotoxins/diuretics when appropriate, albumin challenge, urinalysis, renal ultrasound if obstruction possible |
| Fever or pain        | SBP, cholangitis, abscess, pneumonia, UTI, bacteremia                                                | Cultures, paracentesis, imaging as indicated, early antibiotics                                                       |

## 12. Common Pitfalls

- Assuming normal or mildly elevated AST/ALT excludes cirrhosis. Advanced cirrhosis may have only modest enzyme abnormalities because little functional tissue remains.
- Forgetting diagnostic paracentesis in new or hospitalized ascites. PMN  $\geq 250$  cells/mm<sup>3</sup> changes management immediately.
- Treating hepatic encephalopathy without searching for precipitating GI bleed, infection, constipation, hypokalemia, sedatives, renal failure, or dehydration.
- Biopsying a suspected hemangioma. Vascular lesions can bleed; diagnosis is usually imaging-based.
- Relying on AFP alone to diagnose or exclude HCC. Multiphasic imaging in an at-risk patient is central.
- Overtransfusing variceal hemorrhage without considering portal pressure. Resuscitation is critical, but excessive transfusion may worsen portal pressures in selected patients.
- Missing Wilson disease in a young patient with acute liver failure, neuropsychiatric findings, hemolysis, or unexpectedly low alkaline phosphatase.

## 13. Practice Questions

**1. A 42-year-old with no known liver disease develops confusion, INR 2.1, AST 5700, ALT 6400, and mild bilirubin elevation after several days of taking extra acetaminophen for back pain. What is the immediate disease frame and treatment priority?**

**Answer:** This is acute liver failure from suspected acetaminophen toxicity. The priority is stabilization, glucose monitoring, N-acetylcysteine, evaluation for transplant center transfer, and serial mental status/INR/renal monitoring.

**2. A patient with cirrhosis and new ascites has ascitic albumin 0.8 g/dL and serum albumin 2.4 g/dL. PMN count is 90 cells/mm<sup>3</sup>. What does the SAAG suggest?**

**Answer:** SAAG = 2.4 - 0.8 = 1.6 g/dL, which supports portal-hypertensive ascites. PMN is below 250, so SBP is not diagnosed by cell count.

**3. A cirrhotic patient admitted for melena becomes confused. Name four triggers for hepatic encephalopathy to search for immediately.**

**Answer:** GI bleeding, infection/SBP, constipation, renal failure, hypokalemia, dehydration/excess diuretics, sedatives/opioids, and electrolyte disturbances are key triggers.

**4. A 24-year-old has hepatitis, tremor, depression, low ceruloplasmin, and corneal rings. What disease connects these findings?**

**Answer:** Wilson disease links hepatic injury, basal ganglia/neuropsychiatric findings, low ceruloplasmin, and Kayser-Fleischer rings due to copper accumulation.

**5. An asymptomatic young woman on estrogen therapy has a 6 cm hepatic adenoma. Why is this not handled like focal nodular hyperplasia?**

**Answer:** Hepatic adenomas can rupture and have small malignant potential, especially when large or high-risk. Management includes stopping hormonal exposure and considering resection or other definitive therapy for lesions often above 5 cm.

**6. A patient with cirrhosis has a new 2.5 cm liver lesion. AFP is normal. What is the next diagnostic concept?**

**Answer:** Normal AFP does not exclude HCC. In an at-risk patient, multiphasic CT or MRI looking for characteristic vascular enhancement/washout is the next key diagnostic step.

## 14. Graph-RAG Extraction Map

The following map is designed for graph construction. Each concept should become a node or edge candidate. Edges should preserve directionality when possible: cause -> mechanism -> clinical finding -> diagnostic test -> management -> complication.

| Source concept           | Relationship                   | Target concept                                             | Why it matters                                                                      |
|--------------------------|--------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Cirrhosis                | causes                         | Portal hypertension                                        | Fibrosis and nodules increase intrahepatic resistance                               |
| Portal hypertension      | causes                         | Ascites                                                    | Splanchnic vasodilation, renal sodium retention, and increased hydrostatic pressure |
| Ascites                  | diagnosed by                   | Paracentesis                                               | New or hospitalized ascites requires fluid analysis                                 |
| SBP                      | defined by                     | Ascitic PMN $\geq 250$ cells/mm <sup>3</sup>               | Treatment threshold independent of culture positivity                               |
| Variceal hemorrhage      | treated with                   | Octreotide + antibiotics + band ligation                   | Combines portal flow reduction, infection prevention, and endoscopic hemostasis     |
| Hepatic encephalopathy   | precipitated by                | GI bleeding/infection/constipation/renal failure/sedatives | Treating trigger is as important as lactulose                                       |
| Wilson disease           | caused by                      | ATP7B dysfunction                                          | Copper excretion failure injures liver and brain                                    |
| Hemochromatosis          | screened by                    | Transferrin saturation and ferritin                        | Iron overload is treatable before organ injury                                      |
| Hemangioma               | biopsy contraindicated because | Bleeding risk                                              | Vascular lesion should be diagnosed radiographically                                |
| Cirrhosis or chronic HBV | increases risk of              | Hepatocellular carcinoma                                   | Justifies surveillance and multiphasic imaging of lesions                           |

## References Used for Medical Cross-Check

- American Association for the Study of Liver Diseases (AASLD). Management of Acute Liver Failure practice guidance and educational updates.
- AASLD. Diagnosis, Evaluation, and Management of Ascites, Spontaneous Bacterial Peritonitis, and Hepatorenal Syndrome practice guidance.
- AASLD. Portal hypertension and variceal hemorrhage guidance resources.
- AASLD Practice Guidance on prevention, diagnosis, and treatment of hepatocellular carcinoma, 2023.
- Uploaded Medvale coverage source: gi6.pdf pages on acute liver failure, cirrhosis, Wilson disease, hemochromatosis, benign liver tumors, and hepatocellular carcinoma.